

DRMOE: Towards Better Mixture of Experts via Dual Routing Strategy

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Abstract—Large Language Models are widely utilized in natural language processing tasks but often necessitate expensive fine-tuning to accommodate specific domains. Parameter-efficient fine-tuning (PEFT) techniques help mitigate these costs by reducing the magnitude of trainable parameters. Meanwhile, some recent studies have employed Multi-Task Learning (MTL) to address issues of data scarcity and imbalance in PEFT, demonstrating its effectiveness in overcoming these challenges. The Mixture of Experts (MoE) approach further enhances knowledge sharing in MTL by dynamically assigning experts to different tasks. However, existing MoE approaches face challenges in accurately routing experts due to suboptimal routing methods and random routing. To address these limitations, we propose the Dual Routing Strategy for Mixture of Experts (DRMOE). DRMOE proposes a Dual Routing Strategy to refine expert allocation and constructs a Task-Specified Loss to alleviate randomness in routing decisions. Extensive experiments show that DRMOE effectively enhances performance across tasks, demonstrating its robustness in multi-task scenarios. The code is available at <https://github.com/lhyscau/DRMOE>.

Index Terms—Parameter-efficient Fine-tuning, Multi-Task Learning, Mixture of Experts, Dual Routing Strategy, Task-Specified Loss

I. INTRODUCTION

Large Language Models (LLMs) like ChatGPT [1], [2], Bloom [3] and Llama [4] have advanced rapidly in the field of deep learning, excelling in various natural language processing tasks. Despite their versatility, they often underperform in specific domains and need further fine-tuning. This process is computationally expensive due to the large scale of parameters. Currently, Parameter-Efficient Fine-Tuning (PEFT) techniques [5], such as adapters [6], Low-Rank Adaptation (LoRA) [7], adamix [8], and prefix tuning [9], address this by adjusting only a subset of parameters. These methods reduce computational costs while maintaining performance, making fine-tuning feasible in resource-limited settings.

Furthermore, in cases where data samples are limited or imbalanced, PEFT techniques may fall short of achieving optimal performance. In such circumstances, the role of Multi-Task Learning (MTL) [10] becomes particularly crucial. MTL enables models to learn concurrently from multiple related

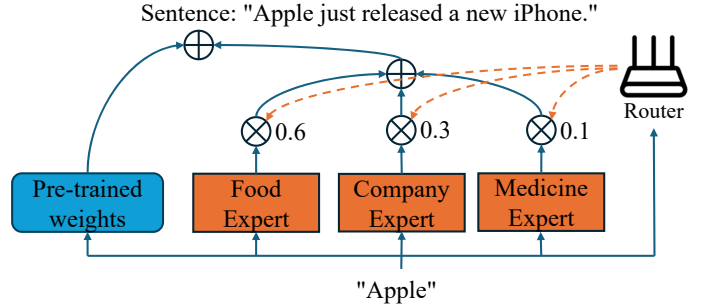


Fig. 1. A motivation for token-level router, which ignore contextual information and result in significant deviations in expert weight contributions.

tasks, thereby facilitating knowledge sharing across different tasks and leveraging shared representations more effectively. Recent studies on leveraging PEFT methods for MTL, such as MoLA [11] and MOELoRA [12], have investigated combining LoRA with the Mixture of Experts (MoE) approach [13]. This combination aims to effectively transfer useful knowledge from related tasks while minimizing interference from unrelated tasks [14].

However, current token-level router methods, such as MoLA, have limitations in practical applications. Specifically, the meaning of a token can vary significantly across sentences, leading to substantial semantic shifts based on context. This variability can affect the accuracy of the routing mechanism. As shown in Fig. 1, in the sentence "Apple just released a new iPhone", the term "Apple" clearly refers to the company, not the fruit. However, suppose the model tended to associate "Apple" more with the fruit during previous training. In that case, the model may incorrectly assign a higher weight to the "food" expert, leading to inaccurate expert weight allocation. This limitation highlights the difficulty of capturing dynamic token semantics across diverse contexts, adversely affecting MoE's performance. Moreover, random routing [15] in MoE models, which randomly assigns inputs to experts often results in training converging on a few experts, thereby causing inefficient use of available experts and limiting the model's ability to fully leverage input information for optimal performance. Enhancing the router's ability to comprehend and adapt to varying semantics for more effective expert allocation

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and reducing the issue of training convergence on only a few experts due to random routing remain significant challenges in the field.

To tackle the aforementioned challenges, this paper proposes a new parameter-efficient approach for MTL called the Dual Routing Strategy for Mixture of Experts (DRMOE). Firstly, we design a Sample Router that utilizes sentence semantics to improve routing, overcoming the limitations of previous token-level routers. Secondly, to further enhance the precision of routing decisions, we introduce a Dual Routing Strategy that cohesively integrates a few Sample Routers with a Task Router, effectively leveraging task category and sentence semantics to optimize the allocation of experts. Finally, an auxiliary loss function is proposed to reduce the impact of random routing by ensuring a balanced token distribution across experts, stabilizing training and improving overall model efficiency. We conducted extensive experiments to demonstrate the effectiveness of DRMOE.

In summary, the contributions of this paper are as follows:

- We design a new routing strategy to capture multi-scale information for expert usage.
- We propose an auxiliary loss function to reduce the impact of random routing and improve the efficiency of expert utilization.
- We conducted extensive experiments on a multi-task dataset, demonstrating the effectiveness of our proposed method.

At the end of this chapter, we outline the structure of the remaining sections of this paper. Section 2 introduces the preliminary background, providing essential concepts and related work. Section 3 presents our proposed method in detail. Section 4 describes the experimental setup and results, demonstrating the effectiveness of our approach. Finally, Section 5 concludes the paper with a summary of our contributions and future directions.

II. PRELIMINARIES

A. Multi Task Learning

Multi-Task Learning (MTL) [10] addresses challenges like overfitting and data scarcity by enabling knowledge transfer across tasks, allowing models to learn shared and complementary features. In LLMs, MTL enhances performance by jointly training on diverse tasks, leveraging shared patterns to generalize better to unseen tasks.

Through instruction tuning, MTL transforms tasks into a unified format, enabling LLMs to excel in few-shot and zero-shot settings. It also reduces the need for extensive fine-tuning by facilitating efficient knowledge transfer, making it a key technique in modern NLP research.

B. LoRA

Low-Rank Adaptation (LoRA) [7] is a PEFT method designed to reduce computational costs while maintaining performance close to full-parameter fine-tuning. The core idea is that weight updates during model adaptation can be effectively captured with a low "intrinsic rank." LoRA achieves this by

decomposing the parameter update for a given pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$ into the product of two low-rank matrices, enabling a more efficient fine-tuning process. The formula for LoRA is as follows:

$$h = W_0x + \Delta Wx = W_0x + BAx \quad (1)$$

where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and the rank $r \ll \min(d, k)$. During training, W_0 is frozen, and only A and B are updated. Here, h denotes the output of the current layer. By updating fewer parameters, LoRA significantly reduces fine-tuning costs.

C. MoE

A MoE module [13], [16] consists of a gate module G and multiple expert modules $E_0, E_1, \dots, E_N$. The gate module dynamically selects and combines outputs from the experts based on the input x . MoE can be expressed as follows:

$$y = \sum_{i=1}^N G(x)_i E_i(x). \quad (2)$$

The gate function $G(x)$ uses a linear layer followed by a Softmax activation function to determine the importance of each expert:

$$G(x) = \text{Softmax}(W_g x) \quad (3)$$

where W_g is a trainable matrix. MoE provides a flexible mechanism for adjusting the contributions of different experts based on different inputs x , enhancing the model's adaptability and performance.

D. Task Router

The Task Router, first introduced in MOELoRA [12], is specifically designed for MTL. MOELoRA integrates the Mixture of Experts (MoE) approach with LoRA, where LoRA serves as an Expert within the MoE architecture. Task Router effectively captures shared features among all samples within a task, providing a robust routing mechanism. Task Router utilizes a trainable task embedding matrix, $\mathbf{E} \in \mathbb{R}^{|T| \times d_T}$, where d_T is the dimension of the task embedding and $|T|$ is the number of tasks. For a given task τ_j , the j -th column of $\mathbf{E}$ is extracted as the task representation vector, denoted $e_j \in \mathbb{R}^{d_T}$. Then e_j is multiplied by a learnable router to derive the global expert weight. Task Router can be expressed as follows:

$$\omega_T^j = \text{Softmax}(\mathbf{W}_T e_j) \quad (4)$$

where, $\omega_T^j \in \mathbb{R}^N$ represents the contribution weight vector tailored for task τ_j . The transformation matrix is denoted as $\mathbf{W}_T \in \mathbb{R}^{N \times d_T}$. Finally, the Softmax operation is employed to scale the contribution weights, serving as the task weights for the experts.

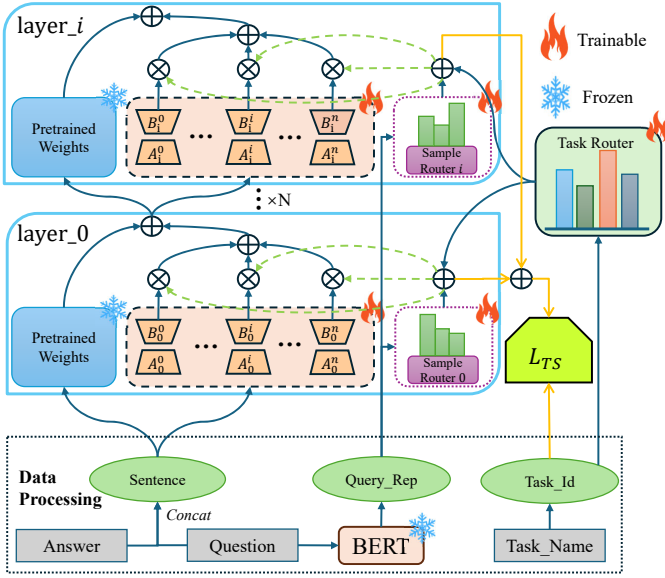


Fig. 2. The overview of data preprocess and training using DRMOE. L_{TS} denotes Task-Specified Loss. In addition, the Answer and Question are concatenated in the format "[Question:Answer]". The fire icon indicates that the module is trainable, while the snowflake icon represents that the parameters of the corresponding module are frozen.

III. PROPOSED METHOD

We propose a novel method, DRMOE, which introduces a Dual Routing Strategy consisting of two types of routers: the Task Router and the Sample Router. Additionally, we design a Task-Specified Loss function to mitigate the effects of random routing, enhancing the model's performance. The data processing component, highlighted within the dashed box, plays a critical role in the workflow and will be elaborated upon in the experiment section. Fig. 2 provides an overview of the DRMOE framework, illustrating its key components and interactions.

A. Sample Router

Previous representative MoE methods in MTL, such as MoLA [11], have faced challenges due to the variability in token meanings across different contexts. This variability can lead to inaccuracies in routing decisions when using token-level routers.

To address this issue, we designed a new module called the Sample Router. As showcased in Fig. 2, the Sample Router within each model layer is independent. We uniformly pass the processed Query_Rep to the Sample Router at every layer. As the hidden representations of experts vary across layers, employing an independent Sample Router for each layer enables the model to extract deeper semantic information and optimize the contribution of each expert based on the specific characteristics of each layer. Since Query_Rep captures the semantics of the entire sentence rather than individual tokens, the proposed Sample Router can avoid the inaccuracies in routing decisions often encountered with token-level routers to some extent.

In the implementation, the Sample Router leverages a sequential linear projection (SLP) mechanism, where two linear layers are systematically combined to achieve enhanced linear representation capabilities and better approximate complex data distributions. Initially, the Query_Rep is processed through the first linear projection to refine its representation. The resulting refined representation is then passed through a second linear projection. Finally, the output is processed with a Softmax function to obtain the final expert allocation weight. Sample Router can be expressed as follows:

$$\omega_{Si}^l = \text{Softmax}(\mathbf{R}_S^l(\mathbf{Q}_i)) \quad (5)$$

$$\mathbf{R}_S^l = \mathbf{W}_2^l \times \mathbf{W}_1^l \quad (6)$$

where, $\mathbf{Q}_i \in \mathbb{R}^{d_h}$ represents the Query_Rep of sample i , and d_h denotes the hidden size of BERT. $\mathbf{R}_S^l$ represents the Sample Router at layer l . $\mathbf{W}_1^l$ and $\mathbf{W}_2^l$ correspond to the first and second layers of the SLP, respectively, with $\mathbf{W}_1^l \in \mathbb{R}^{d_h \times d_p}$ and $\mathbf{W}_2^l \in \mathbb{R}^{d_p \times |T|}$. $|T|$ is the number of tasks.

B. Dual Routing Strategy

The Task Router described in the Preliminary section effectively captures the common features shared by all samples within a task. However, it lacks flexibility. Different samples within the same task may have unique routing preferences, which the current approach does not accommodate.

To address this issue, we propose a Dual Routing Strategy. This strategy captures task-level features and incorporates sample-specific information for routing, providing a more comprehensive basis for routing decisions within the model. Fig. 2 provides a visual representation of our DR, specifically highlighting the area above the dashed box. The strategy integrates the weights of a global Task Router [12] with those from the Sample Router at each layer to compute the final expert weights for each layer. The Task Router generates expert contribution weights based on Task_Id, the identifier for the task category, while the Sample Router generates them using Query_Rep, which represents sentence semantics. By combining two different types of routers, the Dual Routing Strategy enables a more fine-grained MoE. This approach enhances the model's ability to handle diverse and complex input data effectively. The Dual Routing Strategy is expressed as follows:

$$\omega_i^l = \text{Softmax}(\omega_T^j + \omega_{Si}^l) \quad (7)$$

where ω_i^l is the final routing result of layer l , ω_T^j is the Task Router output for task j , and ω_{Si}^l is the Sample Router output for layer l . j is the task ID of sample i .

C. Task-Specified Loss

Previous MoE routing involves high randomness, causing training to often converge on only a few experts, which leads to inefficient use of the available experts [17], [18]. To address this issue, we designed an auxiliary loss function named Task-Specified Loss (TSL). This loss function is based on the idea that each task dataset has a primary expert, while other experts

serve as complementary knowledge sources beyond the main expert. As shown in Fig. 2, the TSL is the green hexagonal shape on the right side. We first collect the routing results from each transformer layer, which is the result of the Task Router and Sample Router. We then aggregate the routing results across all layers and use this aggregated sum, along with the Task_Id, as input variables for the loss function. The formula for TSL is as follows:

$$\mathcal{L}_{\text{TSL}} = \lambda \text{CrossEntropy}(\tau_i, \text{Softmax}(\omega_i)) \quad (8)$$

$$\omega_i = \sum_{l=0}^{N-1} \omega_i^l \quad (9)$$

where λ is a constant used to scale the magnitude of the loss, and N is the number of layers in the model. Meanwhile, ω_i refers to the sum of routing results at each layer. τ_i represents the task ID of the i -th sample. Therefore, this loss function helps to utilize different experts effectively.

IV. EXPERIMENTS

A. Datasets

Our experiments are based on the PromptCBLUE dataset [19], a recently released multi-task Chinese dataset. This dataset consists of 16 diverse medical tasks, each converted into an instruction-following plain text format using specific prompts to ensure compatibility with LLMs. The instruction-following format facilitates more effective training of the LLMs by providing clear and structured guidance, thereby enhancing their ability to perform the designated tasks accurately. Following previous work [12], we select eight task datasets for our experiments to maintain consistency. The dataset's detailed statistical information and evaluation metrics are summarized in Table I.

B. Data Processing

The part enclosed by the dashed box in Fig. 2 illustrates the data processing method used in DRMOE. During the data processing phase, we utilize the **Question**, **Answer**, and **Task_Name** from the original dataset. We concatenate the Question and Answer sections in the format "[Question: Answer]" to form a Sentence for modeling. Furthermore, we use *bert-base-chinese* model [20] to embed the Question to obtain the sentence semantics representation of each sample, referred to as Query_Rep. Each Task_Name is mapped to a corresponding Task_Id, indicating the task category. Finally, the **Sentence**, **Query_Rep**, and **Task_Id** serve as the inputs to the model.

C. Baselines

To evaluate the effectiveness of our proposed DRMOE, we conduct a comparative analysis against several baseline methods, categorized into three groups:

TABLE I
STATISTICS OF THE MULTI-TASK DATASETS PROMPTCBLUE

Task	Description	Size	Metrics
CMeIE	Name Entity Recognition	2828	Micro-F1
CHIP-CDN	Normalization	2381	Micro-F1
CHIP-CDEE	Attribute Extraction	1562	Micro-F1
CHIP-MDCFNPC	Clinic Entity Discovery	4935	Micro-F1
CHIP-CTC	Text Classification	3622	Macro-F1
KUAKE-QIC	Query Intention	3279	Macro-F1
IMCS-V2-MRG	Report Generation	1799	RougeL
MedDG	Dialog Generation	4964	RougeL

1) *LLMs without fine-tuning*: This group includes ChatGPT and Huatuo.

- ChatGPT is evaluated using few-shot learning [21], selecting 3 to 10 input-output pairs from training data to construct the maximum input length.
- Huatuo [22], derived from a Chinese medical instruction dataset, follows the same in-context learning approach [23].

2) *Training strategies with LoRA*: This group encompasses LoRA (Full) and LoRA (Single).

- LoRA (Single) involves traditional LoRA fine-tuning separately for each task without incorporating MTL.
- LoRA (Full) merges the datasets from all tasks into a unified dataset, followed by LoRA fine-tuning on this consolidated dataset.

3) *MoE with LoRA*: This group includes MoLA [11] and MOELoRA [12], which integrate MoE with LoRA for MTL.

- MoLA [11] utilizes a token-level router for expert routing, enabling dynamic selection of experts based on token-level features. Additionally, it incorporates a load-balancing loss function to mitigate the effects of random routing, ensuring a more balanced distribution of tasks among the experts.
- MOELoRA [12] employs a task-level router, which assigns experts based on task-specific characteristics, thereby enhancing the alignment between tasks and expert utilization.

D. Implementation Details

In this study, we use two 1B-scale models: *bloom-1b4-zh* [3], a multilingual model, and *Llama-3.2-1B-Instruct* [24], which is one of the most widely adopted models in current research. Trainable parameters were added to all the linear layers in the decoder layer of the base model. The input and output sequence lengths were set to 1024 and 196. Models were trained with a batch size of 64 for 25 epochs. For MOELoRA, MoLA, and DRMOE, the number of experts was set to 8, corresponding to the number of tasks. The learning rate for DRMOE was 4e-4, while other baselines followed their original settings.

E. Results and Analysis

We present a comparative evaluation of DRMOE and various baseline methods across multiple datasets in Table

TABLE II

THE OVERALL RESULTS OF COMPETING BASELINES AND DRMOE ON PROMPTCBLUE. THE BOLDDED VALUES INDICATE THE HIGHEST SCORES ACHIEVED FOR EACH DATASET WITHIN THEIR RESPECTIVE BASE MODELS. CHATGPT HAS 175 BILLION PARAMETERS, AND HUATUO HAS 6 BILLION PARAMETERS. ALL OTHER METHODS ARE BASED ON MODELS WITH APPROXIMATELY ONE BILLION PARAMETERS, INCLUDING *bloom-1b4-zh* and *Llama-3.2-1B-Instruct*.

Model	Method	CMeIE	CHIP-CDN	CHIP-CDEE	CHIP-MDCFNPC	CHIP-CTC	KUAKE-QIC	IMCS-V2-MRG	MedDG	Average
GPT	Default	0.3058	0.6069	0.2838	0.5854	0.5253	0.4851	0.3253	0.1361	0.4067
Huatuo	Default	0.1826	0.3610	0.1658	0.3487	0.1909	0.1454	0.2401	0.1308	0.2207
bloom-1b4-zh	LoRA(Full)	0.4449	0.8043	0.5545	0.7668	0.8427	0.7930	0.3741	0.1077	0.5860
	LoRA(Single)	0.4606	0.8209	0.5444	0.7749	0.9000	0.7937	0.3739	0.1114	0.5975
	MoLA	0.4380	0.8152	0.5085	0.7634	0.8518	0.7628	0.3691	0.1112	0.5974
	MOELORA	0.4497	0.8229	0.5545	0.7810	0.8645	0.8563	0.3608	0.1093	0.5999
	DRMOE	0.4675	0.8247	0.5622	0.7813	0.8927	0.8597	0.3771	0.1126	0.6097
Llama-3.2-1B-Instruct	LoRA(Full)	0.4718	0.8327	0.5247	0.7232	0.8418	0.7035	0.3403	0.1104	0.5686
	LoRA(Single)	0.4699	0.8534	0.5227	0.7426	0.8827	0.7081	0.3648	0.1174	0.5827
	MoLA	0.4460	0.7927	0.4874	0.7339	0.7727	0.6970	0.3472	0.1129	0.5487
	MOELORA	0.4662	0.8372	0.5258	0.7455	0.8064	0.7025	0.3656	0.1114	0.5701
	DRMOE	0.4736	0.8571	0.5249	0.7456	0.8655	0.7168	0.3778	0.1133	0.6001

TABLE III

THE TABLE PRESENTS THE EXPERIMENTAL RESULTS OF THE ABLATION STUDY FOR DRMOE.

Method	CMeIE	CHIP-CDN	CHIP-CDEE	CHIP-MDCFNPC	CHIP-CTC	KUAKE-QIC	IMCS-V2-MRG	MedDG	Average
w/o DR&TSL	0.4662	0.8372	0.5258	0.7455	0.8064	0.7025	0.3656	0.1114	0.5701
w/o DR	0.4681	0.8462	0.5279	0.7416	0.8591	0.7078	0.3661	0.1127	0.5787
w/o TSL	0.4717	0.8515	0.5213	0.7445	0.8327	0.7241	0.3747	0.1120	0.5791
DRMOE	0.4736	0.8571	0.5249	0.7456	0.8655	0.7168	0.3778	0.1133	0.6001

II. The empirical results demonstrate that DRMOE achieves state-of-the-art performance. Moreover, across different base models, our method consistently surpasses existing baselines, thereby reinforcing its effectiveness and robustness. Finally, we provide a detailed analysis in the following sections.

1) *LLMs without fine-tuning*: The first group of baseline models, consisting of LLMs without fine-tuning, shows significantly poorer performance than other fine-tuned baseline models and DRMOE. While large models exhibit strong generalization capabilities, their performance is still limited, particularly in specific domains. The result highlights the importance of fine-tuning LLMs to incorporate specialized knowledge for specific tasks.

2) *Training strategies with LoRA*: LoRA (Full), which involves fine-tuning a unified dataset combining multiple tasks, performs worse than LoRA (Single). This result suggests that applying LoRA fine-tuning on a mixed dataset for multiple tasks causes substantial inter-task interference. Consequently, traditional PEFT methods in MTL may fail to enable tasks to leverage complementary knowledge from related tasks effectively and may even introduce inter-task interference, resulting in performance degradation.

3) *MoE with LoRA*: We further evaluate two recent MTL methods that utilize the MoE architecture, where the expert module is LoRA. While MoLA and MOELoRA demonstrate strong performance across multiple tasks, our proposed DRMOE outperforms them, achieving the best overall results. Compared to MoLA, which relies on a token-level router, our method better captures contextual information by utilizing a sentence-level router. Moreover, while MOELoRA relies solely on a task-level router, our approach introduces a novel routing strategy that allows for more comprehensive routing decisions. DRMOE helps to avoid the ambiguity of word meanings across different contexts and utilizes a novel routing strategy to achieve more accurate routing outcomes. These advantages likely contribute to DRMOE’s superior performance in multi-task scenarios.

4) *Performance insights across different tasks*: Across the 8 datasets, LoRA (Single) generally outperforms LoRA (Full) except in low-resource settings like CHIP-CDEE and IMCS-V2-MRG, where LoRA (Full) benefits from knowledge transfer. However, LoRA (Full) suffers from inter-task interference in data-rich tasks. Our method effectively shares knowledge and reduces interference, achieving the best performance

across most tasks and handling data imbalance well. For the MedDG task, ChatGPT and Huatuo likely perform better due to their dialog capabilities and larger model sizes.

F. Ablation Study

To investigate the contributions of each component in DR-MOE, we present the results of our ablation study in Table III. The study was conducted using the *Llama-3.2-1B-Instruct* base model. The variant w/o DR replaces the Dual Routing Strategy with the Task Router, resulting in a noticeable decline in performance compared to DRMOE. This highlights the critical role of the Dual Routing Strategy in routing decision. Additionally, we examine the impact of the Task-Specified Loss by evaluating the variant w/o TSL, which omits this component. The performance degradation observed in this variant underscores the importance of the auxiliary loss in mitigating the effects of random routing and enhancing the efficiency of expert utilization. Furthermore, we assess the combined effect of removing both the Dual Routing Strategy and Task-specified Loss with the variant w/o DR&TSL. This configuration does not employ either component. The results for w/o DR&TSL confirm that both the Dual Routing Strategy and the Task-Specified Loss are essential for achieving the superior performance of DRMOE. Overall, the ablation study validates that each component of DRMOE significantly contributes to its effectiveness and robustness.

G. Resource consumption

Unlike methods like MoLA that use a token-level router, DRMOE utilizes sample-level and task-level routers, computing routing weights once per layer for all tokens in a sample. This design is more efficient than MoLA, which recalculates weights for each token at every time step, leading to a linear increase in computational overhead as the token count grows. Consequently, DRMOE reduces training and inference time and offers a more resource-efficient and cost-effective solution.

V. CONCLUSION

In this paper, we have proposed an effective method for improving the performance of multi-task domain adaptation under data imbalance conditions. Our method, DR-MOE, leverages the Dual Routing Strategy and Task-Specified Loss function to help the router allocate expert contributions more precisely. A series of experiments conducted on the PromptCBLUE benchmark dataset, as shown in Table II, have demonstrated the effectiveness of our method compared to several state-of-the-art baselines. These positive results in several base models confirm the utility and practicality of our approach to knowledge transfer. In the future work, we will further explore the way of combining MoE with LoRA for better knowledge transfer.

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